Deep Learning for Neurodegenerative Disease detection through Spontaneous Speech

Master Thesis

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Roger Esteve

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Advisor: Dr Francisco Javier Hernando Pericas
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Abstract

Speech is an attractive non-invasive biomarker for neurodegenerative disease, but multimodal systems that combine acoustic and textual signals expose many interacting design choices whose individual contributions are rarely isolated. This thesis treats such a system as a composition of largely independent design axes (modality, fusion, audio–text alignment, Wav2Vec layer pooling, layer mixing, and feature extractor) and estimates the contribution of each through controlled ablations across three clinical tasks: Alzheimer's disease classification on ADReSSo, amyloid-β status prediction, and primary progressive aphasia variant classification. Its central methodological contribution is a pair of token-to-token alignment schemes, uniform and proportional, that refine the word-to-token alignment of prior work at the subword level. The strongest per-axis choices are consolidated into a single configuration-driven implementation, yielding a recommended architecture that is competitive across tasks while remaining lightweight and deployable.

1 Introduction

Alzheimer's disease is the most common cause of dementia worldwide and is driven by the progressive accumulation of amyloid-β plaques and neurofibrillary tau tangles, a pathological process that begins years before clinical symptoms become apparent [1]. Identifying individuals who harbour this pathology early is a central goal of current research, yet confirmatory assessment still relies on cerebrospinal fluid assays or molecular neuroimaging, procedures that are expensive, invasive, or capacity constrained. At the syndrome level, primary progressive aphasia (PPA) presents additional diagnostic challenges: three canonical variants are recognised (non-fluent/agrammatic, semantic, and logopenic) [2, 3], but behavioural overlap between them makes clinical differentiation difficult, particularly in early or atypical presentations [4]. Both the need for scalable amyloid screening and the challenge of PPA subtyping motivate the development of non-invasive computational tools that can complement expert judgment.

Spontaneous speech is an attractive candidate for this purpose because it simultaneously engages cognitive planning, lexical access, syntactic construction, motor execution, and prosodic control, all of which may be differentially affected by neurodegenerative pathology. Compared to imaging or fluid biomarkers, voice can be acquired frequently, remotely, and at low marginal cost [5]. Structured elicitation tasks such as the Picnic Scene picture description from the Western Aphasia Battery [6] are already part of routine neuropsychological batteries, and recordings from these tasks can be processed computationally without additional clinical instrumentation.

Early computational approaches to cognitive assessment from speech relied on handcrafted linguistic descriptors and shallow acoustic statistics, demonstrating feasibility but limited generalisation [7]. The shift toward deep learning, particularly self-supervised speech encoders such as Wav2Vec 2.0 [8] and contextual text models from the BERT family [9], has substantially improved representational capacity. Community benchmarks such as the ADReSSo challenge [10] have enabled systematic comparison under controlled conditions, and surveys of this literature consistently report that multimodal systems combining acoustic and textual information outperform either modality alone [11].

Despite this progress, most published pipelines report end-to-end accuracy figures that conflate several design choices simultaneously: the modality mix, the fusion architecture, the strategy for aligning acoustic frames with text tokens, and the configuration of the self-supervised encoder. Because these choices are typically entangled within a single system, it is difficult to determine which technique drives the observed performance and which is incidental to the particular implementation or dataset. Furthermore, the majority of benchmarked systems target English corpora, and it is unclear how well observed trends transfer to languages with different morphology and prosodic norms. Spanish-speaking clinical cohorts with pathology-confirmed labels exist through initiatives like the Sant Pau Initiative on Neurodegeneration (SPIN) [12], but these have received limited attention in the multimodal deep learning literature.

1.1 Motivation and objectives

This thesis is motivated by the observation that, although multimodal speech-based systems for cognitive assessment have shown promising results, the field lacks a systematic understanding of which individual design choices drive performance. Published systems vary simultaneously in modality, fusion mechanism, audio–text alignment strategy, and encoder configuration, making it impossible to attribute gains to any single technique. A structured ablation programme, in which each axis is varied while the others are held at controlled defaults, is needed to disentangle these contributions and guide future system design.

Preliminary investigations into deep learning approaches for AD and PPA classification on the ADReSSo and SPIN cohorts were reported in a companion study [13]; the present thesis extends that work by decomposing the pipeline into individually ablatable design choices. The main objectives of this thesis are:

1. **Modality.** Quantify the individual and combined contribution of audio, text, and handcrafted acoustic features across three clinical speech tasks: AD classification (ADReSSo, English), amyloid-β status prediction (SPIN, Spanish, validation-only), and PPA variant classification (the same SPIN cohort, WAB picture-description subset, Spanish, validation-only).

2. **Fusion.** Systematically compare seven sequence-to-sequence fusion families (self-attention, multi-head self-attention, transformer with positional encoding, cross-attention, gated cross-attention, bidirectional cross-attention, and a Mamba state-space comparator), on all three tasks, taking bidirectional cross-attention as the CogniAlign default and replicating the trends under a second, independent implementation.

3. **Alignment.** Propose and evaluate two novel token-to-token audio–text alignment schemes (uniform and proportional) that operate at the subword level, benchmarked against the word-to-token alignment of CogniAlign [14]; this is the central methodological contribution of the thesis.

4. **Wav2Vec pooling.** Evaluate final-layer versus weighted multi-layer Wav2Vec pooling, and an adaptive per-utterance *sample* layer mixer evaluated in the unified implementation.

5. **Feature extractor.** Conduct a single-factor encoder ablation that swaps one audio or text backbone at a time while holding the rest of the pipeline fixed, spanning 24 runs (four encoder variants $\times$ three tasks $\times$ two seeds).

6. **Recommended architecture.** Synthesise the strongest per-axis choices into a recommended hybrid architecture, developed in the Discussion.

The experimental programme traverses these axes as a staged sequence of experiments: a modality study (Step A), a fusion-family comparison (Step B), an alignment study that compares the three alignment schemes under a fixed fusion at the final Wav2Vec layer (Tier A), a Wav2Vec pooling study contrasting the final layer with a learnable weighted sum (WvLayer), and a combined study crossing the alignment schemes with weighted

Wav2Vec pooling (Align×Wv). The strongest per-axis choices are then consolidated and validated in the unified, configuration-driven `res_model` implementation. Throughout, the second implementation (SER) serves only to check that the fusion and Wav2Vec-pooling trends reproduce under an independent software stack (it has no alignment axis), not as a system to be beaten; the unit of comparison is the technique, not the codebase.

The remainder of this thesis is organised as follows. Section 2 reviews the state of the art at the intersection of neurodegenerative-disease biomarkers, clinical linguistics, and multimodal deep learning on spontaneous speech. Section 3 formalises the prediction problem as a composition of six design axes, describes the three clinical cohorts and their preprocessing, the system architecture and its three implementations, the training protocol, and the evaluation metrics. Section 4 reports the staged ablation programme and the unified-model validation, and Section 7 synthesises the findings into the recommended architecture and outlines future work.

1.2 Work plan

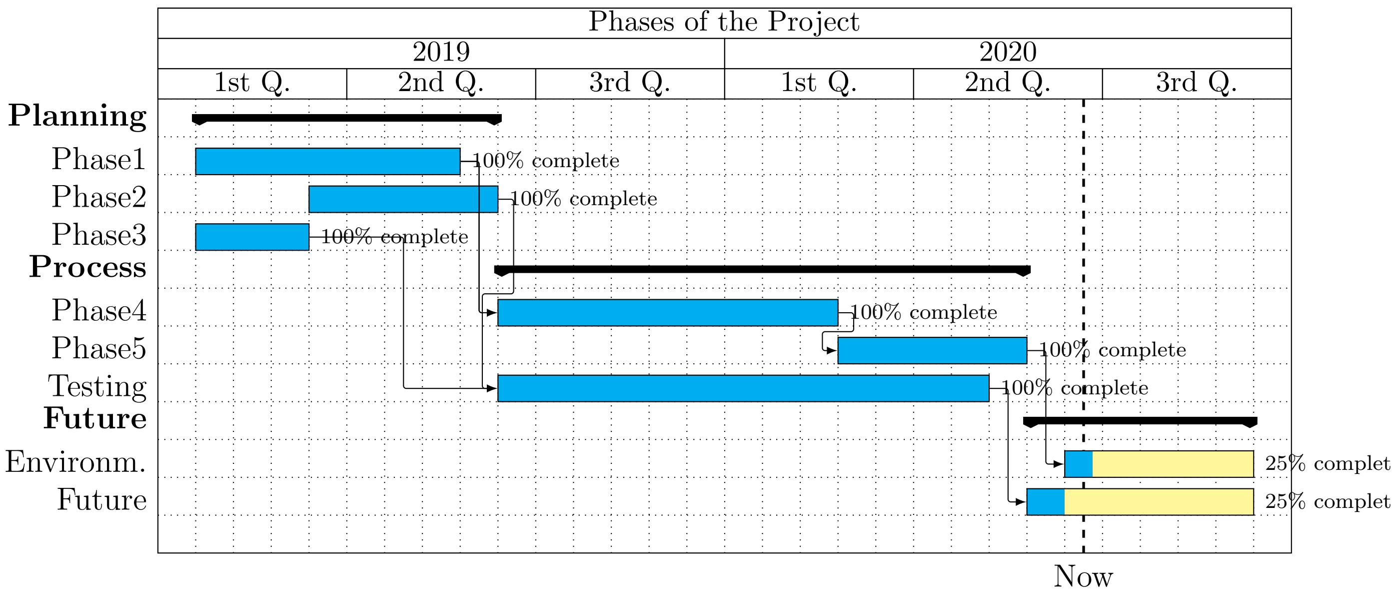

Figure 1: Gantt diagram of the project.

2 State of the art of the technology used or applied in this thesis

This chapter surveys relevant advances at the intersection of neurodegenerative disease biomarkers, clinical linguistics, and multimodal deep learning on spontaneous speech.

This chapter surveys the scientific and technical background that motivates analysis of spontaneous speech with deep multimodal models in neurodegenerative disease. The discussion follows the structure of recent surveys and challenge-led benchmarking in speech-based cognitive assessment [11], but it is organised around the specific threads that underpin the companion contribution developed in this thesis [13]: pathology-aware biomarker estimation from voice, differential diagnosis of primary progressive aphasia (PPA), cross-lingual dataset gaps, self-supervised acoustic encoders paired with language-model text encoders, fusion architectures that align modalities in time, and the complementary role of handcrafted paralinguistic descriptors. Throughout, emphasis is placed on results and modelling trends that are reproducible on public benchmarks, notably the Alzheimer's Dementia Recognition through Spontaneous Speech Only (ADReSSo) corpus associated with the challenge formulation of Luz et al. [10], and on how those trends inform deployment on clinically curated cohorts such as the Sant Pau Initiative on Neurodegeneration (SPIN) [12].

2.1 Biological motivation: amyloid pathology and clinically overlapping syndromes

Accumulation of amyloid-β (Aβ) is a core pathophysiological substrate in Alzheimer's disease (AD) research and underpins current therapeutic strategies that lean on early biological identification of at-risk individuals [1]. Confirmatory assessment traditionally relies on cerebrospinal fluid assays or molecular neuroimaging; although accurate, these modalities are expensive, invasive, or capacity constrained, which contributes to delayed detection relative to symptom onset and limits scalable screening in population settings.

Parallel to biomarker science, differential diagnosis at the syndrome level remains difficult because language and executive deficits arise across neurodegenerative aetiologies with partially overlapping phenotypes. PPA exemplifies this tension: it is defined as a progressive disorder of language as the principal complaint and functional limitation, and contemporary consensus schemes stratify variants according to the dominant cognitive-linguistic profile [2, 3]. The non-fluent/agrammatic (*nfv*), semantic (*sv*), and logopenic (*lv*) variants differ in fluency, grammatical competence, lexical–semantic knowledge, and repetition/spelling-span signatures; importantly, variant labels provide clinically actionable hypotheses about underlying proteinopathy patterns even though symptoms can blur across boundaries [4]. From an engineering viewpoint, these clinical facts motivate pattern recognition systems that are sensitive to subtle temporal micro-structure of speech (planning pauses, articulatory slowing, reduced informational density) and to lexical–semantic content jointly, precisely the multimodal cues accessible from spontaneous picture descriptions or narrative tasks collected during routine assessments [15].

2.2 Spoken language as a scalable digital biomarker

Prospective clinicopathological studies and controlled biomarker programmes increasingly incorporate speech and language markers alongside imaging and fluid assays because voice can be acquired frequently, remotely, and at comparatively low marginal cost [5]. At the same time, evaluation methodology matters: speech measures can track syndrome severity and risk enrichment, yet population heterogeneity (education, dialect, comorbid hearing loss, task instructions) can inflate apparent effects unless modelling assumptions are carefully matched to sampling designs [15].

Early computational work emphasised engineered linguistic descriptors extracted from transcripts (lexical diversity, syntactic complexity, idea density) and shallow acoustic statistics computed from short recordings of spontaneous speech to separate AD from healthy ageing [7]. These approaches demonstrated feasibility but often relied on manual transcription pipelines or handcrafted feature dictionaries whose coverage is incomplete for noisy clinical audio. More recent analyses revisit classical machine-learning baselines (random forests, support-vector machines) with richer descriptors and report uneven sensitivity depending on task elicitation and recording conditions [15]; collectively, such studies argue that marker families carry predictive signal but that shallow decision boundaries under-use temporal dynamics present in the waveform [16, 17, 18].

2.3 From shallow classifiers to deep neural representations

Representation learning replaces brittle front-ends with hierarchical embeddings trained either supervised end-to-end or semi-supervised with objectives aligned to phonetic/lexical structure.

2.3.1 Self-supervised acoustic encoders: Wav2Vec 2.0

In speech processing broadly, self-supervised pre-training of transformer acoustic models learns frame-level features transferable across languages and domains with comparatively modest labelled downstream data. The Wav2Vec 2.0 architecture [8] illustrates this paradigm. A multi-layer convolutional feature encoder first transforms raw waveform samples into a sequence of latent speech representations at approximately 20-millisecond resolution. These latent vectors are then contextualised by a transformer encoder that attends across the full temporal extent of the utterance. During pre-training, a proportion of the latent frames is masked, and the model solves a contrastive task: for each masked position, it must identify the true quantised representation among distractors sampled from the same utterance. Because no transcriptions or labels are required, the procedure scales to vast quantities of unlabelled audio. The resulting transformer layers learn representations at different levels of abstraction: earlier layers tend to encode acoustic and phonetic detail, while later layers capture more abstract prosodic and linguistic patterns. This layered organisation has practical implications for downstream tasks, which may benefit from aggregating information across multiple transformer layers rather than relying on the final output alone, a question addressed experimentally in this thesis. Two configurations of this model are relevant to the present work. The English `base-960h` model, trained on read English speech, is one; the multilingual XLSR-300M variant, which extends the same framework to around 436,000 hours of speech spanning 128 languages, is the other, and its cross-lingual pre-training makes it better suited to scenarios where labelled pathological speech is scarce in any individual language, such as the Spanish cohorts studied here. The thesis employs both as frozen feature extractors and compares them directly in a controlled encoder ablation, since the more powerful multilingual model is not guaranteed to win once the downstream task is fixed. Related self-supervised models, notably HuBERT [19] and WavLM [20], pursue analogous representation objectives through masked-prediction and denoising tasks respectively, but are not employed in this thesis.

[Figure to be added]

Figure 2: Architecture of the Wav2Vec 2.0 self-supervised speech encoder.

2.3.2 Contextual text encoders

On the text side, the BERT family of encoder-only transformer models [9] learns bidirectional contextual embeddings through masked language modelling: random tokens in the input are masked, and the model is trained to predict them from surrounding context. The resulting representations capture lexical semantics and local syntactic structure, going beyond the static word vectors of earlier approaches. Lighter distilled variants such as DistilBERT [21] preserve much of this capacity in a smaller architecture, while modern encoder-only models such as ModernBERT [22] improve representation quality through updated architectural choices. When automatic speech recognition transcripts are available, these text encoders provide a complementary linguistic view that acoustic features alone may encode only implicitly.

Reviews oriented specifically at AD detection from speech highlight two robust tendencies [11]: deep neural pipelines dominate leaderboard-style benchmarks when adequate training partitions exist, and multimodal systems that consume both acoustic embeddings and text outperform either modality alone when transcripts are informative enough for alignment [11]. These observations motivate architectures that treat spontaneous speech as *tightly coupled* audio–text sequences rather than as isolated vectors [23].

2.4 Benchmark challenges and public corpora

Community challenges accelerated reproducible comparison by freezing splits and evaluation protocols for cognitive impairment detection from speech-only submissions [10]. The ADReSSo subset derived from the DementiaBank Pitt corpus of picture descriptions remains the de facto English benchmark for Alzheimer's dementia versus healthy control discrimination under matched recording conditions; numerous challenge systems established competitive baselines [24], and subsequent work continues to extend feature attribution and model explanations [25].

However, performance on English benchmarks does not trivially transfer to other languages: morphology, prosodic norms, and clinician elicitation protocols differ, and labelled neurodegeneration cohorts at scale remain comparatively scarce outside a handful of reference centres. Initiatives such as SPIN address biomarker discovery and validation by assembling multicentre observational data spanning genetics, imaging, fluid markers, and structured cognitive assessments [12]; subsets incorporating spontaneous speech tasks tied to standard batteries (for example picture-description prompts compatible with the Western Aphasia Battery [6]) enable language-specific baselines that complement English challenge corpora.

Spanish-focused clinical studies demonstrate that machine-learning models incorporating language-aware descriptors can achieve strong discrimination between PPA variants [26], while recent bilingual analyses underscore that acoustic models may retain utility even when patients produce speech in a non-dominant language, whereas lexical models become comparatively brittle, a caution worth noting for multimodal pipelines deployed in multilingual clinics [27].

2.5 Automatic differentiation of PPA variants from connected speech

Because syndromic overlap complicates visual inspection alone, quantitative speech analyses seek markers that track breakdown profiles hypothesised by neurolinguistic models: reduced speech rate and grammatical complexity in *nfvPPA*; lexical retrieval failures and semantic thinning in *svPPA*; hesitation and phonological loop limitations with spared grammar in *lvPPA* [2].

Empirical studies highlight pause structure, filler usage, lexical diversity, and discourse coherence metrics as discriminative at group level [28], while articulatory tempo measures help separate non-fluent from fluent variants [29, 30]. At the intersection with amyloid biology, connected speech features associate with cortical amyloid burden within PPA cohorts [18], reinforcing the notion that speech captures not only syndrome labels but correlates of underlying pathology, similar in spirit to speech-only screening models targeting amyloid positivity [16, 17].

[Figure to be added]

Figure 3: Distinguishing speech and language profiles of the three PPA variants.

Recent proof-of-concept studies escalate modelling complexity: large language models combined with curated linguistic measurements attain high classification accuracy in retrospective cohorts [31], whereas neural architectures trained jointly on acoustic and linguistic embeddings yield competitive multi-class accuracy relative to expert benchmarks [32]. Nevertheless, models remain sensitive to sample size and label noise; moreover, cross-cohort validation across accents and tasks remains an open engineering constraint [33].

2.6 Multimodal fusion: pooling, cross-attention, and temporal alignment

Multimodal classifiers must reconcile wildly different sampling rates: acoustic frames at millisecond resolution versus word or sub-word tokens at comparatively coarse timestamps derived from automatic speech recognition [11]. Three broad paradigms describe how audio and text modalities are combined [11]. In early fusion, modality embeddings are pooled along the temporal axis and concatenated into a single fixed-length vector before classification. In late fusion, modality-specific classifiers are trained independently and their outputs are combined at the decision level. Intermediate fusion, the paradigm most relevant to this thesis, processes both modalities jointly through sequence-to-sequence layers that operate on the combined representation before the classification head [13].

[Figure to be added]

Figure 4: Overview of the three multimodal fusion paradigms: early, intermediate, and late.

2.6.1 Self-attention fusion

Self-attention fusion concatenates embeddings from both modalities into a single sequence and processes them with a transformer self-attention layer [34], allowing every position to attend to every other regardless of modality origin. Multi-head variants distribute this computation across parallel subspaces, each potentially capturing different cross-modal interactions. Adding explicit positional encodings (sinusoidal or learned) to the concatenated sequence before applying self-attention provides ordering information that can help preserve temporal structure and modality boundaries within the joint representation.

2.6.2 Cross-attention and gated variants

Cross-attention introduces directionality: one modality supplies the queries while the other provides keys and values, producing representations of the query modality enriched with information from the other. Gated cross-attention extends this by adding a learned gate, typically a sigmoid-activated linear projection, that scales the cross-attended output element-wise before combining it with the original representation; the gate allows the network to suppress uninformative cross-modal contributions on a per-dimension basis. Bidirectional cross-attention runs two cross-attention operations simultaneously (audio queries text, text queries audio) and merges both enriched representations, so that information flows in both directions.

[Figure to be added]

Figure 5: Self-attention fusion over concatenated audio and text embeddings.

2.6.3 Selective state-space models: Mamba

An alternative to attention-based fusion comes from the family of selective state-space models. The Mamba architecture [35] processes the concatenated audio–text sequence through a state-space layer that maintains a compressed hidden state updated recurrently along the time axis. A learned, input-dependent selection mechanism controls how much each position influences the hidden state, enabling the model to retain salient information and discard noise dynamically. Because the state update is recurrent rather than pairwise, computational cost grows linearly with sequence length instead of quadratically as in full self-attention, a property that becomes practically relevant when processing long recordings of spontaneous speech.

2.6.4 Cross-modal alignment

Orthogonal to the choice of fusion mechanism, *cross-modal alignment* addresses the problem of establishing temporal correspondence between streams that operate at fundamentally different resolutions. Acoustic encoders such as Wav2Vec 2.0 produce one contextual frame approximately every 20 milliseconds, whereas text encoders emit one embedding per subword token; a single spoken word may span dozens of acoustic frames while simultaneously being split into multiple subword units by the tokeniser. Without explicit alignment, the fusion layer must learn this correspondence from limited labelled data, risking wasted capacity and spurious associations between acoustically and textually distant positions [14]. The CogniAlign architecture [14] addresses this with a word-to-token alignment strategy. Word-level timestamps are first obtained from the automatic speech recognition step (Whisper), determining which acoustic frames correspond to each spoken word. These word boundaries are then mapped to the subword tokenisation of the text

[Figure to be added]

Figure 6: Cross-attention variants: (a) standard, (b) gated, and (c) bidirectional.

encoder, pairing each textual token with the acoustic frames of the word from which it derives. The aligned representations are subsequently processed by gated cross-attention, where temporal synchronisation encourages the network to attend to co-occurring acoustic evidence (such as jitter, spectral tilt, or prolonged closures) alongside the corresponding lexical content, rather than to correlates from unrelated temporal positions.

This thesis builds on the CogniAlign alignment idea but generalises it along two independent directions. First, where CogniAlign aligns at the word level and attaches every subword token of a word to the same pooled acoustic span, the present work refines the correspondence to the subword granularity itself, introducing two token-to-token schemes: a *uniform* scheme that divides a word's acoustic frames evenly across its subword tokens, and a *proportional* scheme that allocates them in proportion to each token's character length. Second, whereas CogniAlign couples its alignment to a single fusion mechanism (gated cross-attention), this thesis decouples the two and treats fusion as a separate axis spanning the seven families surveyed above, of which gated cross-attention is only one; bidirectional cross-attention is adopted as the default in the resulting grid. Both extensions, and their interaction with Wav2Vec layer pooling, are developed and evaluated in the methodology chapter (Section 3).

Independent multimodal systems repurposed from speaker and emotion modelling demonstrate that competitive fusion architectures transfer across paralinguistic tasks when augmented with domain regularisation [36]; this lineage motivates evaluating robust encoder

[Figure to be added]

Figure 7: Mamba selective state-space fusion applied to a concatenated audio–text sequence.

stacks originally optimised on emotion corpora when ported to clinical speech [13].

2.7 Handcrafted acoustic descriptors alongside learned embeddings

Despite the strength of self-supervised waveform encoders, clinically interpretable descriptors remain attractive because they tie predictions to measurable articulatory and phonatory parameters. These descriptors can be grouped into five broad categories: temporal and pause features (pause histograms, speech rate), prosodic contours (fundamental-frequency variation, intensity envelopes), voice-quality measures (jitter, shimmer, harmonics-to-noise ratio), spectral descriptors (formant tracks, spectral moments), and cepstral coefficients (mel-frequency cepstral coefficients). Pause-oriented statistics repeatedly emerge as salient in cognitive decline [37, 38]; voice-quality perturbations may track neurogenic speech motor control deficits [5].

Open-source dementia-speech pipelines demonstrate competitive accuracy when coupling classical descriptors with modern backbones [25], supporting hybrid designs where auxiliary vectors augment attention pooling or concatenate immediately before classification [13]. Whether explicit features complement latent embeddings in practice depends on architecture and task: pathology-linked cues sometimes persist outside generic waveform embeddings, whereas architectures with explicit alignment may subsume part of the prosodic signal [14].

[Figure to be added]

Figure 8: Cross-modal alignment between acoustic frames and text tokens, comparing unsynchronised input (top) with word-to-token aligned input (bottom).

2.8 Synthesis and positioning

Collectively, the literature supports four design commitments reflected in the methodological arc of this thesis [13]: *(i)* evaluate pathology-aware models where biology-grounded labels exist (for example Aβ status alongside syndrome diagnoses), not solely dementia-vs-control proxies; *(ii)* benchmark competing multimodal fusion philosophies using harmonised preprocessing on both an English reference corpus (ADReSSo) and a clinically curated Spanish-speaking subset aligned with SPIN; *(iii)* extend the word-to-token alignment proposed by CogniAlign [14] with two novel token-to-token schemes (uniform and proportional) and evaluate their interaction with Wav2Vec layer pooling in joint ablations; and *(iv)* quantify when handcrafted acoustic descriptors remain complementary after state-of-the-art self-supervised acoustic encoding and explicit audio–text alignment [11, 25].

The following methodology chapter translates these commitments into concrete architectures, training protocols, and evaluation choices.

[Figure to be added]

Figure 9: Categories of handcrafted acoustic descriptors used in clinical speech analysis.

3 Methodology and Project Development

3.1 Problem Formulation

The general problem addressed in this thesis is the prediction of a clinical label from a single recording of spontaneous speech elicited by a structured picture-description task. Each sample is a pair $(x^{\mathrm{a}}, x^{\mathrm{t}})$, where x^{a} is the raw acoustic waveform and x^{t} is the corresponding transcript of the utterance, and the goal is to learn a mapping $f_\theta : (x^{\mathrm{a}}, x^{\mathrm{t}}) \mapsto y$, where $y \in \mathcal{Y}$ is a discrete diagnostic label whose domain depends on the task.

We study three tasks, each defined over a distinct clinical cohort and summarised in Table 1. The first is Alzheimer's disease classification, a binary discrimination between AD patients and cognitively normal (CN) controls on the English ADReSSo cohort [10], with $\mathcal{Y} = \{\mathrm{AD}, \mathrm{CN}\}$; this is the only task with an official held-out test partition. The second is amyloid-β status prediction, a binary proxy-biomarker task defined on a Spanish-speaking subset of the SPIN cohort [12] elicited with the Western Aphasia Battery picture-description task [6], with $\mathcal{Y} = \{\mathrm{positive}, \mathrm{negative}\}$. The third is primary progressive aphasia variant classification, a three-class problem distinguishing the non-fluent/agrammatic, semantic, and logopenic variants [2] defined on the same SPIN subset, with $|\mathcal{Y}| = 3$. Both Spanish tasks are drawn from this single SPIN subset and share the same data tree; neither has an official external test split, so for these we report cross-validated performance only and never construct an artificial test partition.

Both modalities are encoded by frozen self-supervised models. The acoustic stream is processed by a Wav2Vec 2.0 encoder [8] (the specific variant depends on the implementation and is detailed in Section 3.3) producing a frame-level sequence $\mathbf{H}^{\mathrm{a}} = (\mathbf{h}^{\mathrm{a}}_1, \ldots, \mathbf{h}^{\mathrm{a}}_{T_a})$, and the textual stream by a frozen transformer text encoder of the BERT family [9], producing a token-level sequence $\mathbf{H}^{\mathrm{t}} = (\mathbf{h}^{\mathrm{t}}_1, \ldots, \mathbf{h}^{\mathrm{t}}_{T_t})$. Because the two sequences live on different time bases ($T_a \neq T_t$), an alignment step is required to place acoustic and textual representations on a common index before they can be fused, after which a sequence-to-sequence fusion module combines the aligned streams into a representation that feeds a lightweight classification head.

Rather than treating this pipeline as a single monolithic system optimised end to end, this thesis formulates it as the composition of six largely independent design axes, each of which can be varied while the others are held at controlled defaults. The modality axis m determines whether the classifier consumes audio alone, text alone, their combination, or additionally a set of handcrafted acoustic features. The fusion axis f selects the sequence-to-sequence architecture that combines the aligned acoustic and textual streams, drawn from a catalogue of seven families. The alignment axis a fixes the scheme that maps acoustic frames onto text positions, comprising the word-to-token baseline of CogniAlign [14] and the two token-to-token schemes proposed in this work. The Wav2Vec pooling axis w chooses between the final encoder layer and a learnable weighted sum across layers. The layer-mixing axis ℓ refines that choice: in place of a single global set of layer weights, an adaptive mixer can learn to combine the encoder's layers per utterance, or even per token, so that the depth of representation is selected dynamically rather than fixed in advance. Finally, the feature-extractor axis e varies the pretrained encoders themselves, substituting one audio or text backbone for another, to test how far the downstream trends depend on the specific self-supervised models that supply the two streams.

A concrete system instance is thus parameterised by a tuple (m, f, a, w, ℓ, e), and the central empirical question is to estimate the marginal contribution of each axis (alone and in interaction) to performance on the three tasks, rather than to identify a single best-performing run. The unit of comparison is therefore the *technique*, not the implementation: two independent codebases are used so that the fusion and Wav2Vec-pooling trends can be replicated under a second software stack, with the understanding that cross-stack figures are directional rather than identical reproductions, and a third, configuration-unified implementation then consolidates the resulting techniques into a single model (Section 3.3). Building on this decomposition, the experimental programme of Section 4 traverses the axes in a staged sequence, and the synthesis of the strongest choice along each axis into a recommended hybrid architecture is taken up in the Discussion.

3.2 Datasets and Preprocessing

The three tasks defined in Section 3.1 are evaluated on two clinical cohorts, summarised in Table 1: the English ADReSSo benchmark for AD classification, and a curated subset of the Spanish-speaking SPIN cohort that supplies both the amyloid-β and PPA tasks. They differ in language, label space, and the availability of a held-out test partition, which in turn governs the evaluation protocol applied to each (Section 3.7). All recordings originate from structured picture-description tasks, so the input distribution is comparable across cohorts even though the underlying populations and elicitation languages differ. The cohort descriptions and demographic characterisation below follow the companion paper in which we first introduced this SPIN subset [13]; the remainder of the subsection details each cohort and the common preprocessing pipeline that turns raw recordings into the paired acoustic and textual representations consumed by the model.

Table 1: Clinical cohorts, languages, label spaces, and evaluation protocol. Only the English ADReSSo cohort provides an official held-out test partition; the two Spanish tasks are evaluated by stratified cross-validation. Both Spanish tasks are drawn from the same SPIN subset, elicited with the Western Aphasia Battery picture-description ("Picnic Scene") task.

Task	**Cohort**	**Language**	**Classes**	**Evaluation**
AD vs. CN	ADReSSo	English	2	Official `test-dist` (`test_acc`)
Amyloid-β status	SPIN	Spanish	2	5-fold CV (validation only)
PPA variant	SPIN	Spanish	3	5-fold CV (validation only)

3.2.1 ADReSSo Corpus

The Alzheimer's disease task uses the ADReSSo-IS2021 corpus introduced for the corresponding Interspeech challenge [10]. The corpus is derived from the English-language DementiaBank Pitt Corpus and comprises 237 spontaneous picture-description recordings, totalling approximately 166 minutes of speech, in which participants describe the

Cookie Theft scene from the Boston Diagnostic Aphasia Examination. It is balanced between AD patients and healthy controls and matched for age and sex, which removes the most obvious demographic confounds and makes it a clean benchmark for isolating speech-based effects. The challenge distributes a fixed train and `test-dist` split, so this is the only task in the thesis with an official external test set and therefore the only one on which we report a held-out test accuracy in addition to cross-validated performance. We use ADReSSo primarily to anchor the architecture against a well-documented English benchmark, as in our companion paper [13], rather than to optimise exhaustively for the challenge leaderboard.

3.2.2 SPIN Clinical Cohort

The two Spanish tasks are defined on a curated subset of the Sant Pau Initiative on Neurodegeneration (SPIN) [12], a longitudinal cohort for biomarker discovery and validation in neurodegenerative disorders maintained at the Hospital de la Santa Creu i Sant Pau in Barcelona. From this cohort we assembled a set of Spanish-speaking participants for whom spontaneous speech recordings, a confirmed amyloid-β status, and, where applicable, a clinical PPA variant diagnosis were all available [13]. Unlike ADReSSo, the elicitation stimulus is the "Picnic Scene" from the Western Aphasia Battery [6], and recordings were captured during routine clinical evaluations, with a variable number of samples per subject ranging from one to seven. This repeated-measures structure is the reason all cross-validation on SPIN is stratified by subject identifier, so that recordings from the same participant never straddle the train and validation folds; reporting at the recording level rather than the subject level would otherwise leak speaker identity and inflate the estimated accuracy.

The subset supports two clinically distinct tasks. The amyloid-β prediction task is a binary problem that asks the model to predict amyloid positivity, a fluid biomarker of AD whose confirmation otherwise requires costly or invasive cerebrospinal-fluid assays or PET imaging. At the subject level it comprises 53 amyloid-positive and 49 amyloid-negative participants, and at the recording level 114 positive and 98 negative samples, leaving the task approximately balanced. The PPA variant classification task is a three-class problem distinguishing the logopenic (lvPPA), non-fluent/agrammatic (nfvPPA), and semantic (svPPA) variants, with 35, 22, and 17 subjects and 104, 63, and 35 recordings respectively. This task is markedly imbalanced (svPPA accounts for fewer than a third of the lvPPA recordings), and the small per-class subject counts make fold-level performance estimates high-variance, so the PPA results are interpreted with corresponding caution.

Tables 2 and 3 report the demographic composition of the two tasks. No statistically significant differences in age, sex, or years of education were found across the groups of either task, assessed with the Wilcoxon rank-sum test for continuous variables and Pearson's chi-squared test for categorical ones [13]; this matching is important because it makes it less likely that a classifier exploits demographic shortcuts rather than disease-related speech characteristics.

Table 2: Demographic composition of the SPIN amyloid-β task by subject. Values are mean ± SD or n (%).

	Negative ($N = 49$)	**Positive** ($N = 53$)
Age (years)	72.8 ± 6.1	72.6 ± 6.8
Female	25 (51.0%)	22 (41.5%)
Male	24 (49.0%)	31 (58.5%)
Education (years)	12.8 ± 6.6	11.8 ± 5.7

Table 3: Demographic composition of the SPIN PPA task by subject. Values are mean ± SD or n (%).

	lvPPA ($N = 35$)	**nfvPPA** ($N = 22$)	**svPPA** ($N = 17$)
Age (years)	72.6 ± 6.5	74.7 ± 4.9	73.4 ± 5.8
Female	15 (42.9%)	9 (40.9%)	11 (64.7%)
Male	20 (57.1%)	13 (59.1%)	6 (35.3%)
Education (years)	10.4 ± 6.4	12.6 ± 5.9	9.8 ± 6.9

3.2.3 ASR Pipeline and Transcript Quality

The textual modality is derived from the audio rather than from gold manual transcripts, so that the pipeline reflects a realistic deployment in which only recordings are available. Each recording is first truncated to its initial forty seconds of speech, which approximates the effective input length of the alignment-based model and, on the SPIN data, avoids capturing the clinician speech that occasionally appears toward the end of an evaluation. The truncated audio is then transcribed automatically with Whisper, though the two stacks differ in configuration: the CogniAligned pipeline runs the Whisper `turbo` model offline to obtain transcripts together with the word-level timestamps its alignment step requires, whereas the SER pipeline reuses precomputed transcripts where they are available and falls back to the lightweight Whisper `tiny` model otherwise. The resulting transcript is tokenised by the text encoder before being aligned with the acoustic stream. Because recognition errors propagate into the textual representation, transcript quality is a genuine source of variance, particularly for the Spanish cohort and for the disfluent or aphasic speech characteristic of PPA; the experiments therefore treat the text modality as one ablatable input among several rather than as a noise-free signal.

3.3 System Architecture

The model follows a two-stream design in which audio and text are embedded independently by frozen self-supervised encoders, aligned onto a common sequence index, combined by a sequence-to-sequence fusion module, and finally pooled and mapped to a task label by a lightweight head (Figure 10). Each stage corresponds to one of the design axes introduced in Section 3.1: the encoders and their pooling fix the acoustic and textual representations, the alignment step determines how the two time bases are reconciled, and the fusion module determines how the aligned streams interact. Isolating the stages in this way is precisely what makes the ablation programme possible, since a single axis can be changed while the rest of the pipeline is held fixed.

Both encoders are kept frozen throughout. This is a deliberate modelling choice rather than a computational shortcut: freezing keeps the trainable parameter count in the low millions, concentrates learning capacity in the fusion module under study, reduces the risk of overfitting the small clinical cohorts, and, because the encoder outputs are then deterministic, allows every embedding to be precomputed once and cached, which is what makes a multi-axis grid of experiments tractable. The architecture is realised across three complementary implementations. CogniAligned and SER are the two *primitive* stacks in which the individual techniques were developed and ablated: CogniAligned is the full pipeline and the only stack that implements subword token alignment, whereas SER operates on segment-level representations and serves to check whether the fusion and Wav2Vec-pooling trends reproduce under a second, independent implementation. Because the two stacks differ in their encoders and training schedule, results are compared across them as directional agreement rather than exact reproduction. Building on both, `res_model` is a third, configuration-driven implementation that consolidates these techniques into a single model: any combination of modality, fusion, alignment, Wav2Vec pooling, layer mixing, and feature extractor is selected from one configuration, so that the strongest choices identified separately on the primitive stacks can be assembled and trained together in a controlled way. The two primitive stacks carry the main ablation narrative, while `res_model` consolidates their findings into the unified model from which the recommended architecture is drawn, detailed below. It is competitive with the primitive stacks across the three tasks without being uniformly the best on any single one; its value lies in integration and deployment, not in a higher accuracy ceiling.

3.3.1 Acoustic Encoder: Choice and Configuration

Audio is encoded by a frozen Wav2Vec 2.0 model [8], but the two stacks deliberately use different variants. CogniAligned uses the base configuration (`facebook/wav2vec2-base-960h`): a convolutional feature encoder maps the raw 16 kHz waveform to latent frames at roughly 20 ms resolution (about 50 frames per second), which a twelve-layer transformer context network then contextualises into 768-dimensional hidden states. This model has on the order of 95 million parameters and was trained on 960 hours of English read speech (LibriSpeech); it is therefore monolingual, and CogniAligned applies it unchanged to the Spanish cohorts as well. SER instead uses the cross-lingual XLSR-300M variant, a deeper encoder of roughly 300 million parameters whose twenty-four transformer layers produce 1024-dimensional states and which was pretrained on about 436,000 hours of unlabelled speech spanning 128 languages. The two audio encoders thus differ in depth, width, capacity, and language coverage, which is one of the reasons cross-stack figures are read as directional agreement rather than exact reproduction.

3.3.2 Wav2Vec Pooling and Layer Mixing

The encoder is used purely as a frozen feature extractor: its weights are never updated, and the frame-level sequence $\mathbf{H}^{a}$ is read directly from the transformer stack. How that

[Figure to be added]

Figure 10: End-to-end architecture used in this thesis: frozen Wav2Vec 2.0 and BERT-family encoders produce acoustic and textual sequences, which are placed on a common index by the alignment step, combined by the sequence-to-sequence fusion module, and mapped to a task label by a lightweight head.

sequence is formed is the Wav2Vec-pooling axis. The default, used as the controlled setting in every phase except the dedicated pooling study, takes the hidden states of the final transformer layer. The alternative learns one scalar weight per layer, normalises the weights with a softmax, and forms the representation as the weighted sum of all layer outputs (twelve layers for the base encoder in CogniAligned, twenty-four for XLSR-300M in SER). This is motivated by the well-documented observation that the layers of a self-supervised speech encoder specialise in different information (lower layers carry phonetic and speaker detail while higher layers carry more abstract content), so a learnable combination can expose cues that any single fixed layer discards. To further regularise training on the limited clinical data, the SER stack additionally applies on-the-fly acoustic data augmentation, including additive noise, reverberation, and speed perturbation, prior to feature extraction.

The weighted sum just described shares a single set of layer weights across every recording. Making that combination learnable has a clear purpose: instead of committing in advance to the final layer, the model decides for itself which layers to read features from, putting weight wherever the task-relevant signal is strongest. In the recommended base configuration the weighting is a learnable softmax over all twelve transformer layers, and over all twenty-four for the deeper XLSR-300M encoder.

The most informative depth, however, need not be the same for every recording. A refinement evaluated in the unified `res_model` implementation therefore makes the weighting adaptive, replacing the one fixed vector with a learnable *layer mixer* that forms the softmax over the layer logits conditionally. It comes in three granularities. A *global* mixer keeps one static weighting for all data. A *sample* mixer predicts a separate weighting per utterance, letting an atypical or noisy recording lean on different layers. A *token* mixer goes finer still, mixing the layers independently at each position. A low temperature ($\tau = 0.1$) sharpens the softmax so the mixer settles on a few layers rather than smearing weight across all of them, and its parameters are trained at about ten times the rate of the rest of the head. The sample mixer is the one carried into the recommended configuration: lightweight, input-conditioned, and in practice biased toward the upper layers.

Mechanically, the mixer acts only on the audio branch and only before fusion, collapsing the layer axis into a single vector per time step; the text embeddings never pass through it. The sample mixer obtains its per-utterance weights by mean-pooling each layer over time and passing the result through a small linear scorer followed by the softmax, so the weighting reflects global properties of the recording; the token mixer instead scores every layer at each position, which amounts to a lightweight attention over layers. This layer mixing should not be confused with attention pooling: the mixer reduces the *depth* (layer) axis before fusion, whereas attention pooling would reduce the *time* axis after fusion, and the recommended configuration uses simple mean pooling for the latter.

3.3.3 Text Encoder: Choice and Configuration

Text is encoded by a frozen transformer from the BERT family [9], which turns the tokenised transcript into a sequence of contextual token-level embeddings $\mathbf{H}^{t}$. The two implementations deliberately use different members of this family. CogniAligned uses DistilBERT [21], a six-layer encoder distilled from BERT-base that retains most of its language-understanding capability at roughly half the parameters and inference cost, matching the original CogniAlign design. SER uses ModernBERT-base [22], a more recent encoder of about 149 million parameters trained on the order of two trillion tokens, with a native 8192-token context window, rotary positional embeddings, GeGLU activations, and an alternating local–global attention pattern that makes long-context encoding efficient.

As with the acoustic encoder, the text encoder is frozen and its embeddings are precomputed. Holding it fixed within each stack ensures that the text representation is not a confound when fusion, alignment, or pooling is varied; the difference between the two stacks' text encoders is, however, one of the reasons cross-stack comparisons are interpreted as directional rather than as a controlled head-to-head.

3.3.4 Feature-Extractor Stacks

The choice of pretrained backbone for each stream is itself a design decision rather than a fixed given, and constitutes the feature-extractor axis introduced in Section 3.1. The two primitive stacks already pair different encoders: CogniAligned combines DistilBERT [21] with the English Wav2Vec 2.0 base model, while SER combines ModernBERT [22] with the cross-lingual XLSR-300M. These stacks, however, also differ in their ASR, alignment granularity, and training schedule, so reading the gap between them as an encoder comparison would confound the encoder with everything else that varies. A clean comparison requires changing one encoder at a time while holding the rest of the pipeline fixed.

This is what the unified `res_model` implementation provides through a single-factor extractor ablation. Against a fixed recipe (bidirectional cross-attention fusion, word-to-token alignment, the weighted Wav2Vec stack with the per-utterance sample layer mixer, and word-timestamped CogniAligned transcripts), one component is swapped per run and every other factor is held constant. On the audio side the base encoder is replaced by XLSR-300M, to test whether a deeper cross-lingual model helps the Spanish cohorts; on the text side DistilBERT is replaced by ModernBERT and by the undistilled BERT-base [9], contrasting a lightweight distilled encoder, a recent long-context one, and the original reference. Each variant is run on all three tasks under two random seeds to expose seed sensitivity. The candidates are deliberately drawn from different points of the design space: DistilBERT for its low cost and its role in the original CogniAlign design, ModernBERT for its modern architecture and long context, BERT-base as the undistilled baseline, and the English base versus cross-lingual XLSR audio encoders for the monolingual-versus-multilingual contrast that is most relevant once the same pipeline is applied to Spanish speech. The per-task outcomes, which prove markedly task-dependent, are reported with the experiments.

3.3.5 Cross-Modal Alignment

Before the two streams can be fused they must be placed on a common sequence index, because the acoustic sequence is far longer than the token sequence ($T_a \gg T_t$). CogniAligned resolves this by precomputing an aligned tensor in which each text position is paired with a pooled acoustic representation, and the three alignment schemes differ in how acoustic frames are assigned to text positions. The word-to-token baseline, inherited from CogniAlign [14], uses the word-level timestamps produced by the ASR step to mean-pool the acoustic frames spanned by each spoken word into a single vector, which is then replicated at every subword token of that word. The two schemes proposed in this thesis operate one level finer, at the subword granularity produced by the text tokeniser. The *uniform* scheme takes the frame span of each word and divides it evenly among the word's subword tokens, so that a word split into three tokens contributes three equal slices. The *proportional* scheme instead divides the span in proportion to the character length of each subword token, on the assumption that longer subwords tend to occupy more of the spoken word and should therefore receive proportionally more acoustic frames. Both subword schemes yield a finer-grained audio–text correspondence than the word-level baseline, and the comparison among the three constitutes the alignment axis (Figure 11). SER does not align at the subword level: it fuses segment-level representations and therefore takes no part in the alignment ablation.

3.3.6 Fusion Module

Once the streams are aligned (or, in SER, taken at the segment level), the fusion module combines them into the joint representation that the head consumes. This is the axis the thesis varies most widely. Seven families are implemented and compared, summarised in Table 4, and they fall into three groups that embody three distinct hypotheses about how acoustic and textual evidence ought to interact.

The first group applies **self-attention** to the concatenated audio–text sequence, treating

[Figure to be added]

Figure 11: Audio–text alignment schemes compared in this thesis: the CogniAlign word-to-token baseline, and the two proposed token-to-token schemes (uniform, which spreads each word's frames evenly across its subword tokens, and proportional, which allocates frames to tokens in proportion to their character length).

both modalities as one undifferentiated set of tokens in which every position may attend to every other. Its premise is that the relevant cross-modal structure will surface on its own once the model is given global context, with no architectural separation between the streams. Three variants probe how much machinery that premise actually needs. Plain self-attention uses a single attention map. Multi-head self-attention splits the computation into several parallel heads, each free to specialise in a different relation, one head might track how prosody aligns with content words while another tracks how pauses fall around syntactic boundaries. The transformer-with-positional-encoding variant goes further and injects sinusoidal position information inside a full transformer block [34], so that the order of frames and tokens is modelled explicitly. That last addition matters for speech in particular, where timing is itself diagnostic and a permutation-invariant model would throw away exactly the rhythm that distinguishes fluent from disfluent production.

The second group, **cross-attention**, keeps the modalities apart and lets one query the other. One stream forms the queries while the other supplies the keys and values, so the fused representation is essentially one modality re-expressed in terms of the most relevant parts of the other. Gated cross-attention extends this with a learned gate that scales the cross-modal contribution, giving the model an explicit means to down-weight the other stream when it is unreliable. That control is not cosmetic here: the text stream is the output of an error-prone ASR step, and a fixed cross-modal coupling would force the model to attend to transcription noise it would be better off ignoring. Bidirectional cross-attention performs the exchange in both directions at once, refining the audio with the text and the text with the audio instead of designating either as the primary stream.

The final family abandons attention altogether. The **Mamba** block [35] is a selective state-space model that mixes the sequence in linear rather than quadratic time, and it carries a very different inductive bias: information is propagated through a recurrent hidden state whose dynamics are themselves input-dependent, which favours smoothly evolving temporal structure over the all-pairs comparison that attention performs. Including it asks whether any gain attributed to fusion is specific to attention or survives under a fundamentally different sequence model.

The same seven families are implemented in both codebases. Fixing the catalogue this way is what lets the fusion axis be read as a property of the technique itself, reproducible across two independent software stacks, rather than an artefact of one implementation.

Table 4: The seven sequence-to-sequence fusion families compared in the experiments, grouped by mechanism.

Group	Fusion family	Core mechanism
Self-attention	Self-attention	Single attention map over the concatenated sequence
	Multi-head self-attention	Parallel attention heads over the concatenated sequence
	Transformer + PE	Transformer block with explicit positional encoding
Cross-attention	Cross-attention	One modality queries the other
	Gated cross-attention	Cross-attention with a learned gate on the cross-modal signal
	Bidirectional cross-attention	Cross-attention exchanged in both directions
State-space	Mamba	Selective state-space mixing in linear time

3.3.7 Handcrafted Feature Extraction and Integration

In addition to the learned embeddings, the pipeline can incorporate a set of 55 utterance-level handcrafted acoustic descriptors designed to capture fine-grained paralinguistic cues that are clinically interpretable but not easily recovered by a self-supervised encoder operating on the raw waveform. They are produced by a feature-extraction script adapted from NeuroXVocal [25], which demonstrated the set's effectiveness on the ADReSSo benchmark, and were extended to the SPIN cohort in our companion paper [13]. Extraction relies on `librosa` together with Praat-based analysis via `parselmouth`; the resulting vector is computed once per recording and stored, so that the identical 55-feature set is reused across every experiment that enables it. The descriptors span the five groups listed in Table 5: temporal and pause statistics, acoustic–prosodic measures, voice-quality measures, spectral descriptors, and cepstral coefficients.

Table 5: The five groups of handcrafted acoustic descriptors (55 features in total), extracted with `librosa` and Praat (`parselmouth`).

Group	**Descriptors**
Temporal & pause	Total speech duration, speech-to-pause ratio, number of pauses, pause-duration statistics (mean, maximum, SD)
Acoustic–prosodic	Pitch (mean, SD, range), intensity, speaking rate, articulation rate
Voice quality	Local jitter, local shimmer, harmonics-to-noise ratio
Spectral	Spectral-centroid statistics, zero-crossing rate, formants F1–F3 (mean, SD)
Cepstral	First 13 MFCCs (mean, SD)

The five groups are grounded in the clinical phenomenology of the target disorders. Pause behaviour is the clearest case: a higher pause rate, longer pauses, and a lower speech-to-pause ratio are well-established markers of Alzheimer's disease, where they track the lexical-retrieval and speech-planning difficulties that accompany cognitive decline. Voice quality plays a complementary part, with jitter and shimmer registering the subtle phonatory instabilities that can appear before any overt symptom. What makes such descriptors worth extracting separately is that they are explicit and clinically interpretable, and that a self-supervised encoder offers no guarantee of preserving them. Pretrained to recover phonetic and lexical content, the encoder has no objective that obliges it to retain absolute timing or low-level voice-quality statistics, so it may discard precisely the cues a clinician would attend to.

Before the features are used they are normalised, and the normalisation is computed with care to avoid leakage: each dimension is Z-score standardised using only the mean and standard deviation of the training split of the fold in question, never statistics pooled over the whole dataset.

Integration follows a late-fusion design. The normalised 55-dimensional vector is first projected to the model's 768-dimensional hidden width, then combined with the multimodal stream at one of two late points, prepended as an additional token to the sequence that enters the fusion module, or concatenated to the already-pooled fixed-size embedding immediately before the classification head. Either way the descriptors join only after the self-supervised streams have been formed, so the fusion and cross-attention machinery still operate on the learned representations alone and the handcrafted cues act as an interpretable side-channel rather than something the attention has to disentangle from the waveform.

The descriptors are ablated explicitly in the modality study (Step A), where the audio-plus-text-plus-features cell is compared against the audio-plus-text cell so that any contribution of the handcrafted features is measured against an otherwise identical model. Beyond that controlled comparison, the CogniAligned ablation grids keep the 55-feature vector enabled by default whenever the cohort's feature CSV is available, so the fusion, alignment, and pooling studies are run on top of the audio-plus-text-plus-features input rather than with the features removed.

3.3.8 Classification Heads

The fused sequence is reduced to a single fixed-size vector by a pooling operation (mean pooling in CogniAligned and learned attention pooling in SER), and this vector is passed to a small multi-layer perceptron that outputs the class logits, trained with a cross-entropy objective. The head is kept intentionally shallow so that the bulk of the modelling capacity lies in the frozen encoders and the fusion module under study; this keeps the trainable parameter count low and makes the comparison between fusion families fair, since differences in accuracy can then be attributed to the fusion mechanism rather than to head capacity. The output layer follows the task, with two units for the binary AD and amyloid-β tasks and three for PPA variant classification. All three targets in this thesis are categorical, so the head is always a classifier; the same design would extend directly to a regression target (for instance a continuous amyloid burden rather than a binary status) by replacing the final layer with a single linear unit and the cross-entropy objective with a regression loss, a possibility noted here only to delimit the scope of the present work.

3.4 Unified Implementation: `res_model`

The two primitive stacks established the trends of the individual axes, but each carries incidental design decisions, such as a fixed ASR model, alignment granularity, and training schedule, that make some comparisons impossible to isolate within a single stack. To remove these confounds and to assemble the strongest per-axis choices into one system, the techniques are re-implemented in `res_model`, a configuration-driven trainer in which the modality, fusion family, alignment scheme, Wav2Vec pooling and layer mixer, and the audio and text encoders are all selected from a single configuration. Its results are reported by technique rather than by stack identity, since a given configuration no longer corresponds to "CogniAligned" or "SER" but to an explicit combination of choices. This unified implementation serves three purposes: it reproduces the main technique trends under one codebase, so that they no longer depend on cross-stack differences; it enables the controlled single-factor studies that the primitive stacks cannot express cleanly, namely the adaptive layer mixer of Section 3.3.2 and the encoder ablation of Section 3.3.4; and it provides the vehicle for the recommended architecture.

That recommended architecture is the configuration the ablations point to, anchored on the English ADReSSo task. It pairs word-timestamped transcripts and a DistilBERT text encoder with the Wav2Vec 2.0 base audio encoder, reads the audio as a weighted stack of layers combined by the per-utterance sample mixer, aligns the streams word-to-token, fuses them with bidirectional cross-attention, and classifies a mean-pooled representation with a shallow head; the handcrafted acoustic features are left off by default and enabled only as an ablation. The bundle is deliberately literature-aligned: its word-to-token alignment follows CogniAlign, while its bidirectional cross-attention fusion is the strongest family from the fusion grid rather than CogniAlign's own gated cross-attention. It is chosen as the strongest configuration that stays stable across folds and seeds, not the single highest-scoring cell of any one run. As noted above, on the Spanish cohorts some of these choices change: the cross-lingual XLSR-300M audio encoder, in particular, can be preferable to the English base encoder, so the recommended stack is read as task-anchored rather than universal.

3.5 Training Protocol

All models are trained under stratified five-fold cross-validation applied independently to each dataset, so that every fold preserves the class balance of its cohort. The two implementations use schedules tuned to their respective encoders. CogniAligned is optimised with a learning rate of 2×10^{-5} for up to two hundred epochs with early stopping on the validation fold using a patience of fifteen epochs, while SER uses a learning rate of 5×10^{-5} for twenty-five epochs. To separate genuine effects from initialisation noise on these small cohorts, every `res_model` configuration is trained under two fixed random seeds (43 and 3407) and both runs are reported, so that the stability of a result across seeds can be judged alongside its mean. To keep training and inference tractable with frozen encoders, embeddings are precomputed and cached: CogniAligned stores token-aligned tensors as precomputed `.pt` files, and SER maintains an embedding cache. This caching was added to SER after its initial implementation, so all reported inference times correspond to the final cached configuration.

3.6 Ablation Protocol

The experiments isolate the contribution of each design axis by varying one axis at a time while the others are pinned to controlled defaults, so that any change in performance can be attributed to the manipulated axis rather than to incidental differences. In the CogniAligned grids the defaults are the final-layer Wav2Vec representation, word-to-token alignment, and bidirectional cross-attention fusion, with audio-plus-text input and the 55 handcrafted acoustic features kept on whenever the cohort's feature CSV is available; the only cells without features are the Step A modality cells that explicitly isolate audio, text, or their combination. The programme runs as a staged sequence of experiments: a modality study (Step A); a fusion-family comparison (Step B); an alignment study (Tier A) that compares the three alignment schemes under a fixed fusion (bidirectional cross-attention or Mamba) at the final Wav2Vec layer; a Wav2Vec pooling study (WvLayer) contrasting the final layer with a learnable weighted sum; and a combined study (Align×Wv) that crosses the alignment schemes with weighted Wav2Vec pooling at bidirectional cross-attention. The strongest per-axis choices are then consolidated and validated in the unified `res_model` implementation, which additionally carries the single-factor layer-mixer (Section 3.3.2) and encoder (Section 3.3.4) studies that the primitive stacks cannot express cleanly. The second implementation (SER) replicates only the fusion (Step B) and Wav2Vec-pooling (WvLayer) trends and has no alignment axis; because the two stacks differ in encoders, ASR, and training schedule, cross-stack figures are read as directional agreement rather than as a head-to-head comparison, and the unit of analysis throughout is the technique, not the codebase.

3.7 Evaluation Metrics and Baselines

The primary metric is classification accuracy, reported in a manner consistent with each cohort's evaluation protocol. For ADReSSo we report both the mean cross-validated validation accuracy and the accuracy on the official `test-dist` partition, the latter being the headline figure comparable with the challenge literature. For the two Spanish tasks, which have no official test set, we report validation accuracy only, computed as the mean of the per-fold best validation accuracy across the five folds, and we never substitute an artificially held-out split for the missing test partition. Alongside accuracy, per-sample validation and test inference times are recorded to characterise the compute cost of each technique, since some fusion families are substantially heavier than others. The word-to-token alignment of CogniAlign [14] serves as the literature baseline against which the two proposed token-to-token schemes are measured, and challenge-reported systems on ADReSSo provide the external point of comparison for the AD task. Unless stated otherwise, results are reported as point estimates without significance testing, a limitation revisited in the Discussion.

4 Results

This chapter reports the experimental programme defined in Section 3. Results are presented first per task, in the order Alzheimer's disease classification (ADReSSo), amyloid-β status prediction (SPIN), and PPA variant classification (WAB), and then as cross-cutting ablation studies that isolate the contribution of each design axis. For ADReSSo both cross-validated validation accuracy and the official `test-dist` accuracy are reported; for the two Spanish tasks, which have no official test partition, only cross-validated validation accuracy is reported (Section 3.7). All figures are drawn from the experiment result tables and are reported as point estimates without significance testing.

4.1 Alzheimer's Disease Classification (ADReSSo)

This section reports performance on the English ADReSSo benchmark, the only task with an official held-out test partition, against challenge-reported systems as the external point of comparison.

4.2 Amyloid-β Status Prediction (SPIN)

This section reports validation accuracy on the Spanish SPIN amyloid-β task, including the comparison between the word-to-token alignment baseline and the proposed token-to-token schemes.

4.3 PPA Variant Classification (WAB)

This section reports validation accuracy on the three-class PPA variant task, interpreted with caution given the small and imbalanced per-class subject counts.

4.4 Ablation Studies

This section consolidates the per-axis ablations across the three tasks, following the staged sequence of Section 3.6: modality, fusion family, Wav2Vec pooling, the combined alignment-by-pooling study, the adaptive layer mixer, and the single-factor feature-extractor ablation. Each axis is varied while the others are held at their controlled defaults, and the fusion and Wav2Vec-pooling trends are checked for reproduction under the second, independent stack.

4.5 Error Analysis

A detailed error analysis, including per-class confusion matrices for the three-class PPA task and an inspection of misclassified ADReSSo and amyloid cases, is left as future work. The present study reports point-estimate accuracies without confusion-level breakdowns or significance testing; characterising systematic failure modes (for example, confusion between the logopenic and non-fluent PPA variants, or the influence of ASR transcription errors on the text stream) would require the per-sample prediction logs and is identified here as a direction for subsequent analysis rather than reported with fabricated figures.

5 Budget

6 Environment Impact (Optional)

7 Conclusions and Future Development

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Appendices